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1 **Using Natural Language Processing to Track Negative Emotions**
2 **in the Daily Lives of Adolescents**

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39

Abstract

Tracking emotion fluctuations in adolescents' daily lives is important for understanding mood dynamics and identifying early markers of affective disorders. This study examines whether natural language processing (NLP) methods, including large language models (LLMs), can approximate adolescents' concurrent self-reported negative affect (NA) from ecological momentary assessment (EMA) narratives. We compared nomothetic (group-level) and idiographic (individualized) models as well as different NLP techniques for capturing within-person emotion fluctuations. EMA text responses from 97 adolescents (ages 14-18, 77.3% female, 22.7% male, $N_{\text{EMA}}=7,680$) were analyzed. Text features were extracted using dictionary-based approaches, topic modeling, and emotion ratings derived from a LLM (GPT-4o). Random Forest and Elastic Net Regression models predicted NA from text features, and SHAP values were used to assess feature importance. Key findings, interactive visualizations, and model comparisons are available via the website: <https://emotracknlp.streamlit.app/>. Idiographic models combining text features from different NLP approaches performed comparably to nomothetic models and to GPT-4 alone in R^2 (Nomothetic:.06-.11, Idiographic:.06-.10, GPT-4:.06-.10) but yielded lower Root Mean Squared Error (Nomothetic:0.52-0.88, Idiographic:0.47-0.76, GPT-4:0.85-1.36), improving within-person precision. Importantly, there were substantial between-person differences in model performance and predictive linguistic features. When selecting the best-performing model for each participant (e.g., nomothetic or idiographic), significant correlations between predicted and observed emotion scores were found for 90.7–94.8% of participants. Findings suggest that while nomothetic models and LLMs offer scalability, idiographic models provide greater predictive precision with sufficient within-person data. A personalized approach selecting optimal models for each individual may enhance emotion monitoring and inform targeted interventions through contextual text insights.

64 **General Scientific Summaries**

65 This study shows that natural language processing and machine learning methods can predict
66 adolescents' negative emotional states from text responses with reasonable accuracy. Personalized
67 models that adapt to each individual's language performed better than general models, highlighting the
68 potential of tailored approaches for improving emotion monitoring.

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73 Emotions play a fundamental role in human life, serving as essential cues that influence our
74 actions and interactions with the environment (Barrett et al., 2007). Emotional states act as immediate
75 alerts to potential benefits or dangers, driving us towards actions that align with our personal goals and
76 away from potential threats (Frijda, 1988). Research shows that not only the intensity of emotions but
77 also their temporal dynamics can contribute to the development of mood disorders (Fisher, Fatimah, et
78 al., 2025; Fisher, Pidvirny, et al., 2025; Houben & Kuppens, 2020; Kuppens & Verduyn, 2017). This is
79 especially important during adolescence, a developmental period marked by increases in the frequency
80 of negative emotional states, which may contribute to elevated risk for depression (Bailen et al., 2019;
81 Bennik et al., 2014; Hollenstein & Lanteigne, 2018). Developing effective methods to assess daily
82 emotional changes that are specifically suitable for adolescents is crucial. These measurements could
83 identify both acute emotional distress and longer-term maladaptive patterns in emotions, serving two
84 key purposes: enabling timely interventions during periods of high negative affect and facilitating
85 targeted intervention to address maladaptive emotional patterns.

86 Language offers a powerful avenue for measuring emotion, as people convey their emotional
87 states both explicitly and implicitly through their choice of words and their manner of speaking or
88 writing (Davitz, 2013; Pennebaker et al., 2003). Language plays a dual role in the realm of emotions,
89 serving not only as a primary tool for expressing and communicating emotional states but also in
90 actively shaping how emotions are experienced (Jablonka et al., 2012; Lindquist, 2017). Therefore, the
91 current study aims to evaluate whether linguistic patterns in adolescent's descriptions of emotion-
92 eliciting experiences can be used to track within-person fluctuation in emotional states.

93 With the rise of digital communication, people, especially adolescents, now express themselves
94 extensively through text-based interactions, including social media posts, messages, and online
95 discussions. This shift has generated an unprecedented volume of language data, offering a unique
96 opportunity to study adolescents' emotions on a larger scale (Iliev et al., 2015; Mihalcea et al., 2024).
97 Advances in natural language processing (NLP), a multidisciplinary field combining computer science,

98 artificial intelligence, and linguistics, have enabled researchers to analyze these data efficiently and
99 meaningfully (Hirschberg & Manning, 2015; Popel et al., 2020). While NLP has been used in
100 behavioral science for decades, its accessibility and affordability have significantly improved, making
101 it an increasingly popular tool (Feuerriegel et al., 2025). One key advantage of NLP over traditional,
102 manual text analysis is its ability to handle large datasets quickly and efficiently. By leveraging these
103 advancements, NLP methods can identify patterns in language use (e.g., sentiment, tone, and self-
104 referential quality), that can be used to approximate self-reported emotion ratings (Carlier et al., 2022;
105 Kahn et al., 2007; Sun et al., 2020). Importantly, text-based communication provides insights not only
106 into the intensity of emotions but also into the context in which those emotions arise. This contextual
107 information can reveal situational factors contributing to emotional experiences, offering a more
108 nuanced understanding of emotional processes.

109 In recent years, various text analysis tools have emerged (Boyd et al., 2022; Demszky et al.,
110 2023; Eichstaedt et al., 2021; Neuendorf, 2017; Rathje et al., 2024; Tausczik & Pennebaker, 2010; van
111 Loon, 2022), ranging from basic approaches like counting word frequencies to more advanced
112 methods, such as leveraging large language models (LLMs), each with its own set of advantages and
113 limitations. For example, closed-vocabulary programs (i.e., dictionary-based approach)¹ such as the
114 Linguistic Inquiry and Word Count (LIWC; Pennebaker, 2001), use predefined dictionaries to
115 categorize words. While highly interpretable, transparent, and efficient at summarizing concepts, these
116 methods often neglect context, leading to potential misinterpretations (Eichstaedt et al., 2021). LIWC
117 may classify the word *mad* as an indicator of anger, yet in phrases like "*She's mad talented*", it
118 conveys emphasis rather than negativity, illustrating how context-dependent meanings can lead to
119 inaccurate classifications. In contrast, open-vocabulary approaches, such as Latent Dirichlet Allocation

1 ¹ We use the terms "closed-vocabulary" and "dictionary-based" interchangeably to refer to approaches that rely on
2 predefined word categories (e.g., LIWC). These methods contrast with open-vocabulary approaches, which derive features
3 directly from the text data without relying on preset dictionaries, often using machine learning or statistical models to
4 identify relevant patterns.

120 (LDA), and word embedding methods, leverage data-driven techniques to examine a broader spectrum
121 of words and topics (Blei et al., 2003; Griffiths et al., 2007; Sivakumar et al., 2020). These methods are
122 better at capturing nuances, addressing ambiguous word meanings, and are less susceptible to
123 misinterpretations. Limitations of open-vocabulary approaches include the need for more technical
124 expertise, larger datasets, and careful consideration of parameter choices, as well as challenges in
125 interpretability (Eichstaedt et al., 2021; van Loon, 2022).

126 More recently, LLMs like GPT have shown great promise in accurately rating various
127 psychological constructs in text, overcoming many of the constraints of older methods (Rathje et al.,
128 2024). These models can interpret the context of words and have achieved effective results across
129 multiple languages with simple prompts (Feuerriegel et al., 2025). The main limitations of this
130 approach include a lack of transparency in how inferences were generated and difficulties in
131 reproducing results due to their probabilistic nature (Abdurahman et al., 2024). For example, although
132 LLM-based approaches could predict whether a person experiences negative emotions based on social
133 media posts, LLMs could not be used to identify which linguistic cues (such as the use of pronouns)
134 predict this experience (Feuerriegel et al., 2025; Rathje et al., 2024). In the latter case, dictionary-based
135 methods are preferable. Ultimately, closed- and open-vocabulary approaches, along with
136 advancements in LLMs, provide complementary strengths that may significantly enhance our ability to
137 understand psychological states through language. By combining these methods, researchers may be
138 able to leverage their unique advantages while mitigating their limitations, offering a robust alternative
139 to traditional self-report measures assessing emotions.

140 Despite the increasing sophistication of these methods, significant gaps remain in the current
141 literature. Most importantly, existing studies on text-based emotion prediction have focused on
142 exploring between-person differences (nomothetic approach), primarily identifying which text features
143 are associated with individuals experiencing high levels of negative emotions (e.g., Tackman et al.,
144 2019). These approaches aggregate data across participants and aim to pinpoint common linguistic

145 markers, such as the use of first-person pronouns or negative sentiment, that are indicative of
146 heightened emotional distress and depression (Bathina et al., 2021; Funkhouser et al., 2024). However,
147 this method overlooks both between-person differences in how linguistic markers relate to emotions
148 and within-person fluctuations in emotional states. Even when using multilevel models to predict
149 within-person fluctuations in emotions, models assumes that the same linguistic features signal high
150 levels of negative emotions across different individuals (Funkhouser et al., 2024).

151 Such assumptions fail to account for the fact that language is inherently idiosyncratic, with
152 individuals expressing emotions in unique and context-dependent ways. Beck & Jackson (2022)
153 demonstrated the importance of idiographic approaches, which focus on person-specific patterns, by
154 showing that the psychological and situational antecedents that predicted future loneliness varied
155 substantially across participants, with no two individuals showing the same pattern of predictive
156 features. This highlights the need for personalized models that adapt to the distinctive ways people
157 express emotions through text, enabling more accurate predictions tailored to each individual. To
158 address this limitation, this study integrates diverse language-based tools and leverages machine
159 learning to develop personalized models that use written narratives to infer within-person fluctuations
160 in self-reported negative affect.

161 The study's primary goal was to examine whether within-person fluctuations in negative
162 emotions in adolescents' daily lives can be accurately tracked through text analysis. Since individuals
163 express emotions in a variety of linguistic ways, this research aims to develop more personalized
164 approaches to emotion tracking. Specifically, we address two key questions:

- 165 1) Do idiographic (individual-level) models outperform nomothetic (group-level) models in
166 predicting within-person fluctuations in concurrent self-reported emotions?
167 2) Does combining various NLP approaches improve emotion prediction accuracy?

168 We compare three types of Natural Language Processing (NLP) approaches, which, as noted
169 above, have complementary strengths and limitations:

- a) Closed-vocabulary approach (dictionary-based approach: LIWC and VADER)
- b) Open vocabulary (LDA)
- c) GPT-derived ratings

By addressing these questions, the study aims to benchmark the performance of different language-based approaches for predicting self-reported negative affect from EMA narratives while comparing nomothetic and idiographic modeling strategies. This approach allows us to evaluate how different NLP methods perform in capturing within-person variation in self-reported affect and to assess their potential utility for modeling emotional experiences in daily life. In doing so, this work may inform future efforts to develop language-based approaches for monitoring emotional fluctuations and supporting scalable interventions for youth.

Method

Participants

Participants were derived from two larger studies that recruited adolescents with elevated levels of anhedonia, as well as typically developing (non-anhedonic) adolescents. They included 97 English-speaking adolescents aged 14-18 (75 female, 22 male; $M_{age} = 16.2$, $SD = 1.9$) recruited from the greater Boston area from January 2016 to November 2023. Participants were excluded based on a history or current diagnosis of any of the following DSM-5 psychiatric illnesses: schizophrenia spectrum or other psychotic disorder, bipolar disorder, substance or alcohol use disorder within the past 12 months or lifetime severe substance or alcohol use disorder. Participants were also excluded based on current diagnosis of anorexia nervosa or bulimia nervosa or if they had a neurodevelopmental disorder that would interfere with study tasks. Due to the neuroimaging component of this study, additional exclusion criteria included fMRI contraindications. For additional information about sample inclusion and exclusion criteria see Murray et al., (2023).

194 **Procedure**

195 All procedures were approved by the Mass General Brigham IRB. Participants who were 18
196 years of age provided written informed consent; participants who were under 18 provided written
197 assent, with their parents providing written consent. At the baseline session, either in-person or over
198 Zoom, participants were administered a semi-structured clinical interview, the Kiddie Schedule for
199 Affective Disorders and Schizophrenia (K-SADS; Kaufman et al., 1997) and completed self-report
200 measures. Following the baseline session, participants installed the MetricWire app on their
201 smartphones to complete ecological momentary assessments (EMAs). Of the participants, 39
202 completed the EMA for four consecutive weeks; for the first 5 days, surveys were delivered 2-3 times
203 per day in the afternoon and evening, and for the remaining 25 days, surveys were delivered 4 times
204 per day. The other 58 participants completed the EMA in 5-day (Thursday through Monday) blocks
205 every other week, with 2-3 survey prompts per day, for a median of 17 weeks.

206 **Measures**

207 ***Ecological Momentary Assessment (EMA)***

208 Participants were asked to rate on a 5-point Likert scale, ranging from 0, “Very slightly or not at
209 all,” to 4, “Extremely” the extent to which they were feeling several emotions immediately before they
210 started the assessment. Participants’ negative affect (NA) included responses for: “sad,” “nervous,”
211 and “angry.” Mean NA was measured by averaging these three variables. In addition, participants
212 responded to the following open-ended questions: 1) What were you thinking about right before you
213 started this survey? 2) Think about the most enjoyable or happy time since you completed the last
214 survey (or if this is your first survey, then the last 24 hours). Very briefly, what happened (1-2
215 sentences is fine)? 3) Think about the most stressful or negative time since you completed the last
216 survey (or if this is your first survey, then the last 24 hours). Very briefly, what happened (1-2
217 sentences is fine)? Participants with fewer than 30 observations were excluded from the analysis.

218 *Language Measures*

219 We used the following strategies to extract text features and generate quantitative summaries of
220 the language data.

221 **Closed-vocabulary.** For close vocabulary (dictionary-based approach), we extracted features
222 using the Linguistic Inquiry and Word Count (LIWC; Tausczik & Pennebaker, 2010) and the Valence
223 Aware Dictionary and sEntiment Reasoner (VADER; Hutto & Gilbert, 2014). LIWC is a computerized
224 text analysis tool that categorizes words into over 90 linguistic and psychological dimensions based on
225 an internal dictionary of approximately 6,400 words (Boyd & Schwartz, 2021). It calculates the
226 percentage of words that match each predefined category, offering insights into linguistic structures
227 (e.g., pronouns), psychological constructs (e.g., affect), and broader language patterns (e.g., analytical
228 thinking). LIWC has been extensively validated across numerous studies and is widely used in
229 psychological research to quantify language use patterns associated with various psychological states
230 and traits. For this study, we used LIWC-22 (Boyd et al., 2022), the latest version of the software, the
231 analyze participants' responses to the EMA open questions.

232 VADER is a simple rule-based model for general sentiment analysis optimized for social media.
233 It provides four sentiment scores: negative, positive, neutral, and compound (an overall sentiment
234 score from -1 to +1). VADER is particularly effective at handling sentiments expressed in short,
235 informal text and accounts for factors like punctuation, capitalization, and modifiers (e.g., intensifiers
236 like "very") that influence the intensity of the sentiment. Whereas LIWC excels in offering detailed
237 psychological and linguistic insights across a wide range of texts, VADER is more attuned to detecting
238 sentiment polarity and intensity in social media. Finally, text lengths were extracted as features for
239 each question and included in the models.

240 **Open-vocabulary.** Latent Dirichlet Allocation (LDA) is a probabilistic clustering method that
241 identifies topics based on word co-occurrence rather than relying on predefined dictionaries(Blei et al.,
242 2003; Griffiths et al., 2007). This approach allows LDA to group semantically related words while

243 considering context, reducing word sense ambiguity. The latent topics were extracted from the
244 preprocessed text corpus using probabilistic modeling with the R package *topicmodels*(Grün & Hornik,
245 2011). In our study, we implemented LDA using two distinct approaches for nomothetic and
246 idiographic analyses. For the nomothetic model, we extracted a uniform set of topics across the entire
247 sample. For the idiographic model, we applied LDA separately for each individual, identifying topics
248 based on their unique word usage patterns. To interpret the extracted nomothetic topics, **Table 1**
249 presents the 15 most frequent words per topic, offering a semantic representation of the clustering.
250 Details on topic selection, model tuning, and evaluation metrics are provided in the Supplement
251 **Section S1** and **Fig. S1**. Furthermore, **Fig. S2** depicts the distribution of LDA-derived topics across
252 participants, highlighting variations in topic prevalence within the dataset.

253 **Generative Pre-trained Transformer (GPT): LLM-as-Rater Approach.** To generate
254 emotion ratings using large language models (LLMs), we used GPT, a deep learning-based AI
255 language model developed by OpenAI. GPT is built on a transformer-based architecture, a neural
256 network design that excels at processing sequential data by using self-attention mechanisms to capture
257 contextual relationships across words. This allows GPT to generate human-like text by predicting the
258 next word in a sequence based on the provided context. Specifically, we used GPT-4, an advanced
259 version pre-trained on a vast dataset (45TB), enabling it to generate coherent sentences and perform
260 various tasks such as writing, answering questions, and engaging in conversations. In this study, we
261 implemented GPT-4 as an LLM-as-a-rater, a measurement strategy in which the LLM is provided with
262 a structured prompt specifying an explicit rating task and applies that instructed measurement rule to
263 the text. GPT-4 was applied here as a general-purpose rater without any person-specific adaptation.
264 Recent work has begun developing personalized LLMs that tailor outputs to individual users through
265 fine-tuning or retrieval augmentation (Liu et al., 2025). Such approaches were not employed in the
266 current study. We prompted GPT-4 through the application programming interface (API) using Python
267 *openai* package to rate the extent to which a participant experienced one of the following emotions:

268 Sadness, Anger, and Nervousness, on a scale of 1 to 5 (similar to the scale used in EMA), based on
269 responses to three open-text questions. Using the API ensured application of the same standardized
270 prompt, fixed model parameters (e.g., temperature), and consistent, reproducible outputs across the
271 dataset. See **Section S2** in the supplement for the full prompt used to generate GPT responses.

272 **Data Analysis**

273 *Model Specification*

274 We began our analysis by extracting text data from EMA responses and conducting both lexicon-
275 based and transformer-based analyses. Next, we applied preprocessing steps (e.g., text normalization,
276 stopword removal, stemming) to refine the text before performing topic modeling. Finally, we trained
277 machine learning models to predict negative emotions based on the derived text features, evaluated
278 model performance, and determined feature importance. **Figure 1** illustrates this analysis pipeline in
279 detail.

280 We compared both nomothetic (group-level) and idiographic (individual-level) models in their
281 ability to predict variability in negative emotional states *within* individuals over time. We employed
282 two machine learning approaches to predict negative emotions: elastic net regression and random
283 forest models. Elastic net regularization (ENR) is a popular variant of conventional regression that
284 combines two types of penalties: ridge and lasso. This combination helps address issues related to
285 multicollinearity by constraining the coefficients of correlated variables while also minimizing model
286 overfitting. On the other hand, random forest (RF) is an ensemble learning method based on decision
287 trees. Unlike ENR, random forest can capture complex nonlinear relationships and interactions
288 between variables without having to specify them in advance. Given that these two methods rely on
289 different algorithms (i.e., penalized regression vs. decision tree-based approaches) for selecting
290 variables, assessing their importance, and generating predictions, comparing them can help to
291 determine which one provides the most accurate predictions in a given context.

292 To assess model performance and generalizability while minimizing overfitting, we implemented
293 a nested cross-validation (CV) procedure using the *nestedcv* package in R (Lewis et al., 2023). A full
294 description of the nested CV procedure is provided in **section S3** in the Supplement. We evaluated
295 model accuracy using R^2 (the square of the correlation coefficient) and RMSE. R^2 was our primary
296 metric because we wanted to prioritize the model's ability to detect increases in negative emotional
297 states, reflected in corresponding increases in predicted scores. While a stronger correlation (and thus
298 higher R^2) indicates better alignment between predicted and observed values, it does not necessarily
299 imply lower average prediction error, which is captured by the RMSE.

300 Initially, we ran nomothetic models using features extracted from all three NLP approaches,
301 including all subjects for group-level analysis. These models identified common predictors of negative
302 emotional states at the group level. To evaluate how well these group-level models generalized to
303 individual participants, we calculated individual metrics, such as R^2 , derived from the nomothetic
304 models for each participant. Subsequently, we employed fully idiographic models, building separate
305 models for each participant to capture person-specific language-emotion associations. These models
306 focused exclusively on within-person variability, aiming for highly individualized predictions. In these
307 idiographic models, variables with low variance were removed based on standard frequency criteria,
308 where the most common value for that variable could not exceed 95% of the total observations.
309 Finally, we compared the three abovementioned approaches, to understand their relative strengths in
310 predicting negative emotions.

311 NLP feature extraction procedures were consistent across modeling approaches, with the
312 exception of LDA topics, which were estimated at the sample level for the nomothetic analyses and
313 separately for each participant at the idiographic level. GPT ratings and LIWC/VADER features were
314 identical across approaches.

315 ***Feature importance***

316 To evaluate feature importance, we used SHAP (SHapley Additive exPlanations) values, which
317 quantify the contribution of each feature to the model's predictions in a consistent and interpretable
318 manner by assessing how variations in a feature impact the model's output (Lundberg, 2017). SHAP
319 values were calculated using the R packages *fastshap* and *ggbeeswarm*, which facilitates visualization
320 and interpretation of feature contributions. This method allowed us to identify the relative importance
321 of predictors in the model and their specific effects on the predicted outcomes.

322 **Transparency and Openness**

323 Because the data involve sensitive information, and in accordance with IRB requirements, they
324 are available upon reasonable request to the corresponding author (HF) and contingent on a data use
325 agreement. All analysis code and raw output files are available at <https://osf.io/grjy9> (Fisher, 2026).
326 Model results were saved separately for each participant, model type, and emotion, resulting in
327 multiple CSV files organized by model type (nomothetic, idiographic, NLP approaches, ML models).
328 The aggregated values reported in the manuscript (e.g., mean R^2 across participants) were derived
329 directly from these raw outputs using the provided analysis code. We report how we determined our
330 sample size, all data exclusions, all manipulations, and all measures in the study. This study was not
331 preregistered.

332 **Results**

333 The full results can be explored in the web app (<https://emotracknlp.streamlit.app/>), which also
334 includes an interactive chatbot powered by a GPT-based LLM. The chatbot was trained on the full
335 manuscript and is designed to answer questions about the research, figures, and results presented in
336 this paper, as well as additional findings.

337 **Demographic and clinical characteristics**

338 The original dataset included 156 participants. Of these, 57 had fewer than 30 EMA observations
339 and were excluded, leaving 99 participants. Two additional participants showed no within-person
340 variation in their EMA emotion ratings across the entire study and were therefore removed, resulting in

a base analytic sample of 97 participants. For specific emotion analyses, additional exclusions could occur when EMA sessions were removed through listwise deletion due to missing feature values (e.g., LIWC/VADER, LDA topics, or GPT-derived ratings). In some cases, non-zero emotion ratings occurred only in sessions with missing feature values; after listwise deletion, the remaining sessions therefore contained only zero ratings, making model performance undefined for that emotion. These participants were excluded from that specific emotion analysis. In addition, idiographic models used 10-fold cross-validation within each participant. For participants with very few non-zero observations, some test folds could contain only zero ratings, making performance undefined and leading to additional exclusions from idiographic results. As a result, idiographic sample sizes were sometimes smaller than nomothetic ones. **Table 2** presents the demographic and clinical characteristics of the participants. For the participants who completed EMA in four consecutive weeks (see Methods for EMA schedule), the median number of EMA surveys completed was 77 ($M = 75$, $SD = 24$), and mean EMA compliance was 67% ($SD = 21\%$). For the participants who completed EMA in biweekly 5-day blocks, the median number of EMA surveys completed was 66 ($M = 68$, $SD = 24$), and mean EMA compliance was 51% ($SD = 17\%$).

Model performance: Nomothetic Approach

Table 3 presents model performance metrics for predicting negative emotions using combined text features from three NLP techniques. Three modeling approaches are compared: group-level nomothetic (one model combining all observations), per-participant nomothetic (same modeling approach as group-level nomothetic but performance is calculated separately for each individual), and idiographic (separate models for each individual). Further details and results, including separate models for each NLP technique, are available in the Web app (see "Model performance across sample" and "Model performance per participant").

The group-level nomothetic model performed well, with R^2 values ranging from 0.11 to 0.38. However, when evaluated separately for each participant – which is critical given the ultimate goal of

366 tracking fluctuations in emotions *within* individuals – performance decreased substantially. Mean R^2
367 values fell between 0.06 and 0.11, with considerable variability in predictive accuracy across
368 participants. For example, when predicting mean negative affect (NA), individual R^2 values ranged
369 from 0.00 to 0.41, demonstrating that while text features explained little to no variability in NA for
370 some, they accounted for a significant proportion of variability in others. See **Table 3** for results for
371 the specific emotions of sadness, anger, and nervousness.

372 **Model performance: Idiographic Approach**

373 The idiographic approach (separate models for each individual) showed similar average
374 performance to the nomothetic approach, with mean R^2 values ranging from 0.06 to 0.10. However, the
375 idiographic approach demonstrated better performance in terms of root mean squared error (RMSE)
376 (0.47 to 0.76) compared to the nomothetic model (0.52 to 0.88), suggesting lower prediction error.
377 **Figure 2** displays the relationship between predicted and observed (actual) NA ratings using person-
378 specific (idiographic) models. Each colored line represents an individual participant's predicted
379 estimates (y-axis) across different levels of actual NA (x-axis). While the overall trend, represented by
380 the dashed black line, suggests a generally positive relationship between predicted and actual ratings,
381 indicating that the models capture meaningful within-person fluctuations in NA, there were large
382 between-person differences in the strength of these associations.

383 **Figure 3** displays the substantial differences in model accuracy across individuals by
384 showcasing examples of four high- vs low-accuracy person-specific models (figures for all other
385 outcomes and for all participants for both the nomothetic and idiographic models are available on the
386 Web app “True vs. Predicted”).

387 **Examining Between-Person Differences in Which Text Features Predict Negative Emotions**

388 **Figure 4** illustrates the variability in feature importance by showing the top 10 text features from
389 the four best-performing subject-specific random forest models predicting NA. These models are the
390 same four high-accuracy models displayed in **Figure 3**. The results show two types of variability

391 between individuals: differences in model performance and, even among highly accurate models,
392 differences in the text features that contribute to predicting NA. Critically, this visualization
393 demonstrates how text features provide valuable contextual information. For example, one participant
394 (participant 2) exhibits more negative emotions when using words related to their family ("Family"),
395 yet another (participant 3) shows greater negative emotions when focusing on the past ("FocusPast").
396 Similarly, a different participant (participant 4) expresses more negative emotions when writing about
397 work-related topics ("Work") but demonstrates less negativity when using words associated with
398 acquiring objects, states, or goals that fulfill personal needs ("Acquire"). This rich information can be
399 used to create person-specific profiles that enable us to identify the specific contexts in which negative
400 emotions arise, providing insights that can be used to personalize interventions. A heatmap of the
401 feature importance across participants and feature importance figures for each participant, outcome,
402 and ML model are available on the Web app ("Feature importance heatmap" and "Feature importance
403 per participant"). Details on the differences in predictive text features between high- and low-
404 performing models can be found in the Supplement (**Section S4** and **Figure S3**).

405 **Which NLP Approach Performs Best in Predicting Negative Emotions?**

406 **Table 4** and **Figure 5** display the performance metrics for each NLP approach used in the
407 idiographic models. When examining the idiographic predictive performance of each NLP approach
408 separately, GPT generally showed better performance for NA, sadness, anger, and nervousness, with
409 R^2 values around 0.10 for NA and sadness, and slightly lower for anger and nervousness. These
410 results were comparable to those achieved by the idiographic models that combined all NLP
411 approaches (see **Table 2** for the combined model results), except in the prediction of nervousness
412 where GPT slightly outperformed the combined idiographic model. However, it is important to note
413 that while GPT had relatively strong R^2 values, it also had a higher RMSE compared to the
414 combined models, suggesting a higher average prediction error despite accounting for more
415 variance in the outcomes. LIWC+VADER and LDA each demonstrated poorer predictive

performance, with the lowest average R^2 values and fewer significant associations across all emotions, especially for anger and nervousness. This highlights that the contextual understanding provided by GPT was more effective in capturing within-person emotional fluctuations compared to the other methods when NLP approaches were tested individually. Overall, while GPT captures variability in negative emotions (as reflected by R^2), its predictions are less precise in terms of exact values (as indicated by RMSE). Combined models, which balance capturing variability and minimizing prediction errors, offer a more robust and reliable approach to emotion prediction.

Evaluating Best Performing Models for Each Individual

To investigate the effectiveness of personalized models, we identified the best-performing predictive model for each of the 97 participants. Using this flexible approach, the vast majority (90.7-94.8%) of the participants showed a significant correlation between predicted and observed (actual) emotion scores. The table detailing the best model for each participant and each emotion outcome can be found in the Web app under "Best model performance". **Figure 6** shows the distribution of best-performing models across participants (see also in the web app under "Best Model Performance"). Idiographic models were most frequently selected for all outcomes except for Anger, where nomothetic models performed best for 44.3% of the participants. Elastic Net showed the best performance for most participants across outcomes, followed by Random Forest. For NLP approaches, the model combining all text features outperformed for most of the participants (52.6-59.8%). Interestingly, though GPT showed the highest mean R^2 scores, it was chosen as the best model for only a relatively small portion of the participants (11.3-19.6%).

Are Prediction Models Emotion-Specific?

To examine whether our models captured emotion-specific signals rather than general NA, we compared the extent to which each emotion-specific model (both Random Forest and Elastic Net) predicted each observed emotion. Specifically, for each observed emotion outcome (NA, Sad, Angry, and Nervous), we calculated R^2 and RMSE between the observed and the predicted emotion, either

441 matching (e.g., predicted and observed sadness) or mismatching (e.g., predicted anger and observed
442 sadness). **Figures S4-S7** display box plots of the distribution of R^2 and RMSE for each emotion. As
443 shown in the figures, overall, and consistent with the interpretation that the models have emotion-
444 specific signals, the matched predictions showed higher R^2 and lower RMSE than mismatched
445 predictions, indicating that the models capture unique, emotion-specific variance. However, for RMSE,
446 we sometimes observed better performance for NA relative to the specific emotions, suggesting that
447 the signals underlying broad NA might be more robust or reliable relative to single emotion ratings.

448 **Does model performance vary by symptom levels, demographic characteristics or Number of**
449 **Observations?**

450 We conducted exploratory analyses examining whether model performance varied as a function of
451 symptom levels, demographic characteristics, and number of observations (Tables S1 and S2 in the
452 supplement). After FDR correction, only higher anhedonia (SHAPS) was associated with better
453 prediction of sadness in the idiographic random forest model ($\rho = .35$, $p = .001$, $pFDR = .032$). No
454 other associations were significant after correction, including demographic variables and number of
455 observations.

456 **Discussion**

457 In this study, we combined multiple Natural Language Processing (NLP) approaches to examine
458 whether within-person fluctuations in self-reported negative affect can be inferred from EMA
459 narratives using NLP. By comparing different modeling strategies and language-based features, the
460 results provide a benchmark for future research examining language-based inference of affect in EMA
461 contexts. Recognizing the idiosyncratic nature of emotional communication, we compared idiographic
462 models, tailored to individual patterns, with nomothetic models that capture common trends across
463 groups. The use of NLP and machine learning techniques to analyze language may provide new
464 opportunities for studying emotions and the contexts in which they arise in daily life, which could
465 ultimately inform future approaches to mental health assessment and clinical decision-making. The

466 results showed that, overall, group-level nomothetic models showed high performance in continuously
467 predicting negative emotions (R^2 range: 0.11-0.38). These findings align with previous studies
468 demonstrating the utility of NLP approaches in detecting emotional states across participants (Akhtar
469 et al., 2019; Tanana et al., 2021). However, since these studies primarily focused on identifying
470 emotional states at the between-person level, rather than capturing *within-person* fluctuations, their
471 high performance does not necessarily indicate an ability to accurately track moment-to-moment
472 emotion fluctuation within individuals. When we calculated performance metrics for each participant
473 individually, the model performance declined significantly and showed high between-person
474 variability, indicating that the nomothetic models' ability to track within-person changes varies
475 considerably from one individual to another.

476 When we built separate models for each participant (idiographic models) and compared their
477 performance to that of nomothetic we found that the mean R^2 was comparable between the two
478 approaches, however, RMSE was lower for the idiographic models. In addition, idiographic models
479 revealed significant between-person variability in the text features predicting emotional states,
480 indicating that individuals express negative emotions in distinct linguistic ways. This raises an
481 intriguing question: if individuals differ so markedly in their predictive features, how can a nomothetic
482 model that captures only general trends perform just as well, at least in terms of R^2 ? One plausible
483 explanation is a trade-off between statistical power and individual variation. While idiographic models
484 capture unique, person-specific patterns, nomothetic models benefit from larger datasets that enable the
485 estimation of multiple, weaker yet common feature-emotion relationships. In other words, although
486 individual differences exist, the robust common patterns identified by the nomothetic approach appear
487 sufficient to achieve similar predictive accuracy. Supporting this interpretation, idiographic models
488 showed lower RMSE and higher accuracy for more participants compared to the nomothetic models.
489 However, nomothetic models demonstrated greater consistency across participants, with more

490 participants presenting significant associations between predicted and actual negative emotions
491 compared to idiographic models.

492 Rather than directly comparing nomothetic and idiographic models, we also tested an approach
493 that selected the best-performing model for each participant. Using this method, idiographic models
494 were most often the best performers, and the vast majority of participants (90.7–94.8%) showed a
495 significant correlation between predicted and observed emotion scores. A limitation of this approach is
496 that choosing the “winner” per participant may lead to overfitting or capitalize on chance, inflating
497 accuracy estimates. At the same time, the results suggest that some participants may need
498 individualized models to capture their unique patterns of expression, whereas others can be modeled
499 well using data pooled across individuals. Balancing personalization and generalizability is therefore a
500 key challenge for NLP and machine learning approaches to emotion prediction.

501 While we are not aware of studies directly comparing nomothetic and idiographic approaches in
502 text analysis, previous research comparing their effectiveness in tracking mental states using passive
503 sensor data from smartphones and actigraphy, as well as self-reported ecological momentary
504 assessment (EMA), has yielded similar results (Aalbers et al., 2023; Cheung et al., 2017; Rozet et al.,
505 2019; Soyster et al., 2022). For example, Aalbers et al. (2023) used smartphone passive sensor data to
506 predict stress levels. Their findings revealed that idiographic models demonstrated higher accuracy in
507 tracking stress levels for some participants (Spearman’s ρ rank-order correlation up to 1 in idiographic
508 models vs. up to .65 in nomothetic models). However, nomothetic models significantly predicted stress
509 for a larger proportion of participants (up to 23.2% for idiographic models vs. up to 55% for
510 nomothetic models). Rozet et al. (2019) used a comparable method and initially found that nomothetic
511 models performed better (i.e., were more accurate) than idiographic models. However, as more data
512 accumulated, the performance of the idiographic model eventually equaled and then surpassed that of
513 the nomothetic model, suggesting that idiographic models may be a better option when sufficient data
514 is available at the single subject level.

515 When comparing specific NLP approaches, GPT-4, implemented here as an LLM-as-rater,
516 achieved the highest R^2 among the individual models, comparable to the model combining all three
517 NLP approaches. However, GPT-4 also showed higher RMSE than the combined model, suggesting
518 that despite capturing meaningful variance in emotional states, it was less precise in predicting their
519 exact numerical values. One possible contributor to this pattern is that GPT ratings tended to show
520 lower variance than the EMA ratings, which can lead to larger residuals and therefore higher RMSE.
521 Furthermore, when selecting the best-performing model for each participant, GPT was chosen for only
522 a relatively small portion of individuals, indicating that its predictions are relatively stable across
523 individuals but not necessarily the most accurate for any single participant. This pattern aligns with the
524 nomothetic-idiographic trade-off discussed earlier, as GPT operates as a nomothetic model that
525 generalizes across individuals rather than incorporating person-specific information. Future research
526 should aim to optimize this balance by leveraging fine-tuning techniques, such as participant-specific
527 calibration or supervised fine-tuning on emotion-labeled data, to improve GPT model accuracy.

528 From an applied perspective, GPT is already trained on a vast corpus of data (approximately 45
529 terabytes of text data from various sources (OpenAI et al., 2023)) which makes it particularly attractive
530 for studies with limited datasets, as it does not necessarily require retraining on task-specific data to
531 yield meaningful predictions. However, the model's lack of transparency and inability to explicitly
532 articulate the rationale behind its predictions remain critical limitations that researchers must carefully
533 weigh (Feuerriegel et al., 2025). Given these considerations, we recommend that researchers
534 thoughtfully evaluate GPT's role in their analytical pipeline, considering whether to employ it as a
535 standalone tool or integrate it with complementary approaches. Future work should focus on
536 developing methods that combine GPT's powerful inference capabilities with more transparent
537 analytical approaches, potentially offering a more robust framework for emotion analysis in
538 psychological research.

539 Finally, exploratory analyses suggested slightly better prediction among adolescents with higher
540 symptom levels, although these associations were weak and no consistent effects were observed for
541 demographic characteristics. If replicated in larger samples, this pattern may further support the
542 potential relevance of language-based emotion prediction for identifying early markers of affective
543 disorders.

544 **Limitation and future directions**

545 This study is the first to use different NLP approaches to directly compare nomothetic and
546 idiographic models for tracking emotional fluctuations. Yet, findings of this study should be
547 interpreted in light of several limitations. First, capturing the nuanced dynamics of emotional
548 expression typically requires large amounts of data. In everyday interactions, subtle emotional changes
549 may only become apparent with extensive exposure to an individual's language use, much like
550 knowing someone well enables you to discern small shifts in their mood. In our analyses we did not
551 find a significant association between the number of observations and predictive performance. This
552 suggests that, within the range of observations available in the current dataset, differences in sample
553 size were unlikely to account for the observed differences in accuracy. However, it remains possible
554 that substantially larger numbers of observations per participant could improve idiographic model
555 performance, and future studies with longer monitoring periods may be better positioned to evaluate
556 this possibility. Therefore, future studies should strive to collect larger within-person datasets,
557 potentially sourced from daily-life communications, to enhance the ability to track and model these
558 nuances. Second, only one generative LLM was evaluated, and performance may vary across different
559 models or model versions.

560 Third, the overall predictive accuracy was modest, leaving significant room for improvement. In
561 addition to increasing the quantity of data, future research could benefit from incorporating additional
562 modalities (e.g., vocal features, facial expression, passive sensors from smartphones or wearables such
563 as smartwatches or smart rings, which are becoming increasingly popular) alongside text. Prior studies

564 have shown that combining various modalities can enhance the performance of nomothetic models of
565 emotion (Kalateh et al., 2024), and thus this multimodal approach may also improve idiographic
566 models. Moreover, future work should explore whether individuals not only differ in the text features
567 that predict their emotions but also in the types of modalities that best capture their emotional states.
568 For example, individual differences, such as the tendency to use affective suppression, may influence
569 how emotions are conveyed in text, with greater suppression potentially leading to lower model
570 performance. Future studies should examine whether these participants require larger datasets to
571 capture subtle nuances in their text, or if alternative modalities (e.g., passive sensor) better detect
572 changes in their emotional states.

573 Fourth, the current study compared idiographic and nomothetic approaches, demonstrating that
574 each has its merits. Future studies could explore hybrid approaches that integrate both individual-
575 specific and group-level information, balancing personalization with statistical power to enhance
576 predictive accuracy. Finally, the text data in our study consisted of responses to specific questions,
577 which limits the generalizability of our findings to other types of text (e.g., text messages and social
578 media posts). Future research should incorporate text from diverse sources such as social media,
579 conversational exchanges, and free-form writing, to determine whether these results extend to broader
580 contexts.

581 Finally, in this study, negative affect was reported prior to the narrative prompts, reducing the
582 likelihood that recalling and describing emotionally valenced events inflated concurrent affect ratings.
583 However, because two prompts explicitly asked participants to describe stressful or positive
584 experiences, predictive performance likely reflects, at least in part, the emotional focus of the prompts.
585 This is consistent with recent meta-analytic work on depression prediction showing that text type
586 matters, with depression-relevant text (e.g., clinical interviews) yielding stronger predictive
587 performance than more general language in predicting depression (Fisher et al., 2026). Accordingly,
588 the findings should be interpreted within the context of this structured EMA protocol.

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766 **Table 1**

The 15 most frequent words within the six topics extracted using LDA

Topic number	Topic interpretation	
1	Social & Evening Activities: focused on social interactions and nighttime activities with friends	friend, talk, sleep, last, night, last_night, see, time, morning, hang, wake, late, talk_friend, boyfriend, hang_friend,
2	Academic Life: centered on school-related activities and academic responsibilities	homework, school, think, class, finish, test, now, take, tomorrow, math, studi, right, finals, essay, stress
3	Home activities: capturing leisure activities, particularly around media consumption and family time	watch, nothing, eat, dinner, play, game, show, family, movie, tv, ate, video, eat_dinner, watch_tv, favorite, watch_movie
4	Family Interactions: reflecting family relationships and dynamics	mom, think, fun, sister, new, dad, brother, room, read, make, made, book, something, fight, clean
5	Daily Activities: representing routine daily activities like meals	go, went, home, walk, outside, lunch, food, drive, music, back, listen, shop, car, hurt, around, dog,
6	Personal States & Obligations: describing emotional states, needs, and work-related responsibilities	get, work, feel, want, today, day, need, good, think, just, felt, like, done, tired, sick,

767
768

Table 2

Demographic and Clinical Characteristics of the Sample

Sample Characteristics (N = 97)		
	n	%
Biological Sex		
Female	75	77.3
Male	22	22.7
Race		
American Indian	0	0
Asian	13	13.4
Black	10	10.3
Pacific Islander	1	1.0
White	65	67.0
Multiracial	5	5.2
Other race	3	3.1
Ethnicity		
Hispanic	4	4.1
Not Hispanic	92	94.8
Unknown	1	1.0
Current Diagnoses (DSM-5)		
MDD	21	21.6
GAD	17	17.5
SAD	5	5.2
Panic Disorder	3	3.1
Specific Phobia	2	2.1
ADHD	1	1.0
PTSD	3	3.1
Medication		
Psychotropic Medication	8	8.2
Age (years)		
	M (Range)	SD
	16.2	1.9
	(12 – 18)	
Family Income (USD)		
	198,895	111,478
	(0 – 500,000)	

Note. MDD = Major Depressive Disorder; GAD = Generalized Anxiety Disorder; SAD = Social Anxiety Disorder; PTSD = Post-Traumatic Stress Disorder; ADHD = Attention-Deficit/Hyperactivity Disorder.

Table 3
Comparison of Group-level Nomothetic, Per-Participant Nomothetic, and Idiographic Model
Performance for Negative Affect, Sadness, Anger, and Nervousness Using Random Forest and
Elastic Net

Type of Model	Nomothetic Group-Level Performance			Nomothetic Per-Participant Performance				Idiographic			
	R ²	R	RMSE	R ²	R ² range	R	RMSE	R ²	R ² range	R	RMSE
Negative Affect											
Random Forest	.38	.62	0.55	.10 (.10)	.00; .41 Sig.n= 53/97	.26 (.17)	0.52 (0.22)	.10 (.12)	.00; .47 Sig.n= 42/97	.21 (.25)	0.47 (0.20)
Elastic net	.17	.41	0.64	.11 (.09)	.00; .39 Sig.n= 56/97	.28 (.17)	0.61 (0.27)	.10 (.12)	.00; .53 Sig.n= 47/97	.15 (.28)	0.48 (0.20)
Sad											
Random Forest	.33	.58	0.75	.10 (.12)	.00; 0.53 Sig.n= 48/96	.25 (.21)	0.73 (0.27)	.10 (.13)	.00; .46 Sig.n= 38/94	.17 (.26)	0.68 (0.22)
Elastic net	.14	.37	0.85	.09 (.09)	.00; .42 Sig.n= 51/96	.24 (.19)	0.82 (0.32)	.10 (.11)	.00; .51 Sig.n= 46/94	.10 (.30)	0.69 (0.23)
Angry											
Random Forest	.20	.45	.69	.09 (.12)	.00; .69 Sig.n= 37/90	.22 (.20)	0.63 (0.33)	.06 (.07)	.00;.33 Sig.n= 27/85	.08 (.23)	0.64 (0.29)
Elastic net	.13	.36	.72	.07 (.07)	.00; .45 Sig.n= 49/90	.19 (.10)	0.88 (0.32)	.07 (.08)	.00; .43 Sig.n= 39/85	.09 (.27)	0.66 (0.31)
Nervous											
Random Forest	.24	.49	.85	.06 (.08)	.00; .44 Sig.n= 33/95	.17 (.06)	0.81 (0.29)	.08 (.12)	.00; 0.71 Sig.n= 33/94	.10 (.27)	0.75 (0.27)
Elastic net	.11	.33	.92	.07 (.07)	.00; .35 Sig.n= 42/95	.18 (.07)	0.88 (0.33)	.09 (.12)	.00; 0.84 Sig.n= 43/94	.12 (.30)	0.76 (0.29)

Note. Sig. *n* represents the number of participants for whom the association between actual ratings and model-predicted ratings was statistically significant.

Table 4

Comparison of Idiographic Model Performance for Negative Affect, Sadness, Anger, and Nervousness Using LIWC+VADER, LDA, and GPT

	LIWC+VADER				LDA				GPT			
	R ²	R ² range	R	RMSE	R ²	R ² range	R	RMSE	R ²	R ² range	R	RMSE
Negative Affect												
Random Forest	.05 (.08)	.00;.42 Sig.n=31/97	.19 (.24)	0.49 (0.21)	.07 (.09)	.00;.48 Sig.n=36/97	.11 (.23)	0.49 (0.21)	.10 (.08)	.00;.34 Sig.n=55/97	.26 (.18)	0.85 (.20)
Elastic net	.07 (.10)	.00;.55 Sig.n=36/97	.08 (.30)	0.49 (0.22)	.06 (.07)	.00;.44 Sig.n=35/97	-.12 (.21)	0.50 (0.22)				
Sad												
Random Forest	.05 (.08)	.00;.38 Sig.n=24/95	.15 (.25)	0.70 (0.25)	.05 (.08)	.00;.46 Sig.n=25/95	.08 (.21)	0.71 (0.26)	.10 (.10)	.00;.54 Sig.n=55/96	.25 (.19)	1.11 (0.26)
Elastic net	.07 (.08)	.00;.44 Sig.n=42/95	.02 (.30)	0.70 (0.26)	.07 (.07)	.00;.29 Sig.n=45/95	-.17 (.20)	0.71 (0.27)				
Angry												
Random Forest	.04 (.06)	.00;.28 Sig.n=15/70	.06 (.20)	0.72 (0.29)	.03 (.05)	.00;.58 Sig.n=12/85	.03 (.18)	0.68 (0.32)	.09 (.10)	.00;.44 Sig.n=45/90	.22 (.21)	0.93 (0.34)
Elastic net	.05 (.07)	.00;.27 Sig.n=26/70	-.06 (.25)	0.73 (0.31)	.07 (.07)	.00;.28 Sig.n=43/84	-.21 (.17)	0.67 (0.32)				
Nervous												
Random Forest	.04 (0.06)	.00;.29 Sig.n=19/86	.10 (.24)	0.80 (0.26)	.05 (.08)	.00;.53 Sig.n=29/94	.10 (.20)	0.77 (0.28)	.06 (.06)	.00;.29 Sig.n=40/95	.18 (.18)	1.36 (0.22)
Elastic net	.05 (.07)	.00;.35 Sig.n=25/86	.00 (.28)	0.81 (0.28)	.05 (.05)	.00;.23 Sig.n=36/94	-.15 (.17)	0.76 (0.28)				

Note. Sig. n represents the number of participants for whom the association between actual ratings and model-predicted ratings was statistically significant.

781 **Figure 1**

782 The pipeline for processing and analyzing text data from EMA responses to predict emotional states

783

784 *Note.* The process begins with preprocessing (e.g., text normalization, stopword removal,
785 stemming). Text features are then extracted using lexicon-based analysis (LIWC, VADER),
786 transformer models (GPT-4), and topic modeling (LDA). These features, including GPT emotion
787 ratings, linguistic categories, sentiment scores, and topics, are fed into machine learning models
788 (Random Forest, Elastic Net) for prediction. Model performance is evaluated using R^2 and
789 RMSE, while SHAP values provide insight into feature importance. This approach integrates
790 multiple NLP techniques to enhance emotion prediction accuracy.

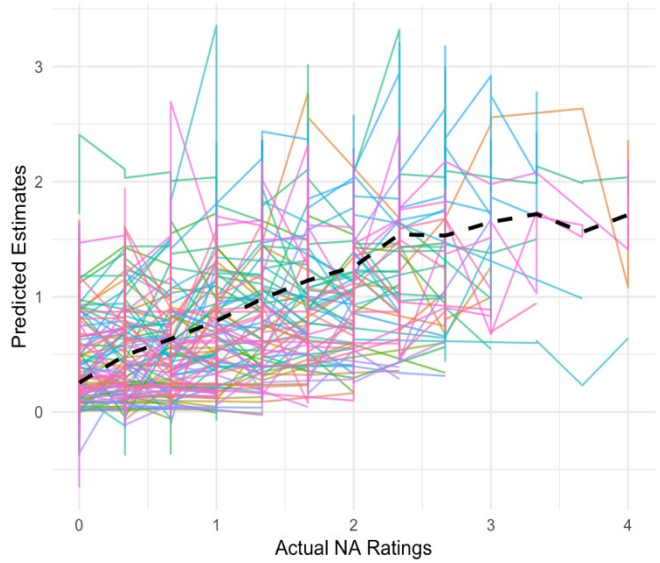
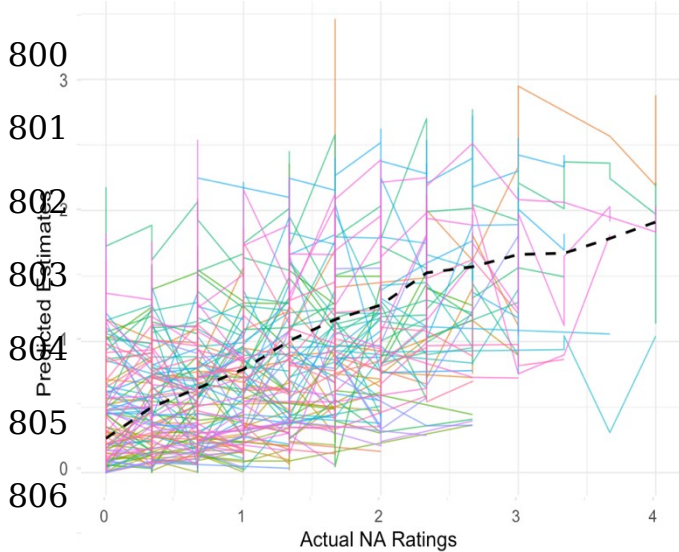
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Figure 2.

The relationship between predicted and observed (actual) negative affect ratings using person-specific (idiographic) models

Predicted vs Actual Negative Affect Ratings

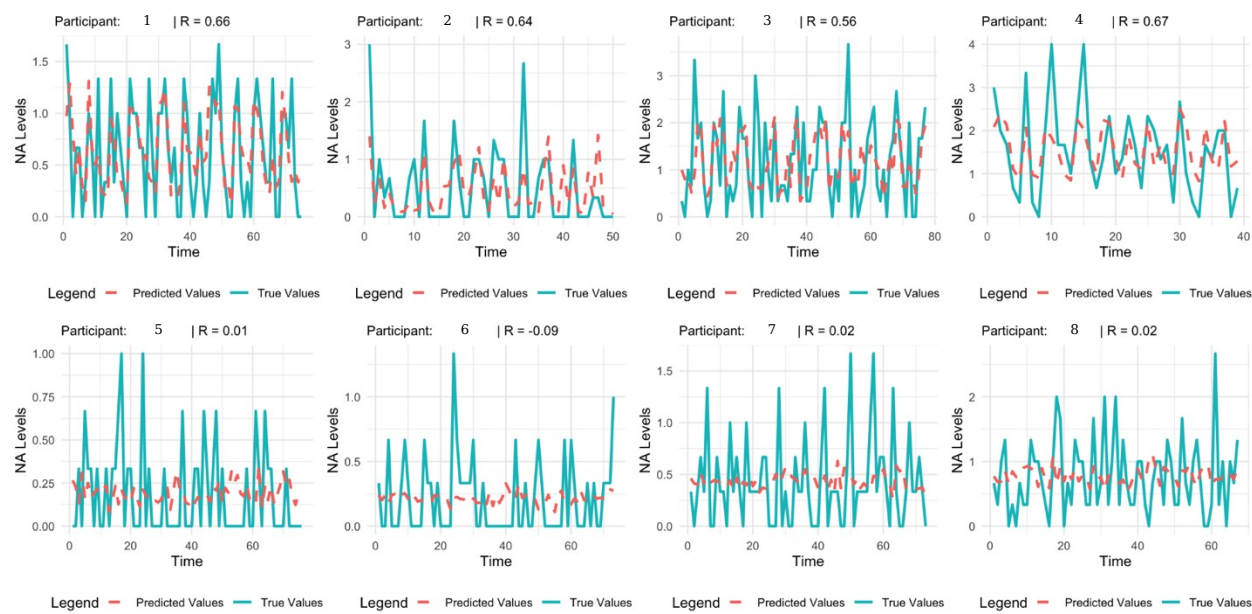
Predicted vs Actual Negative Affect Ratings



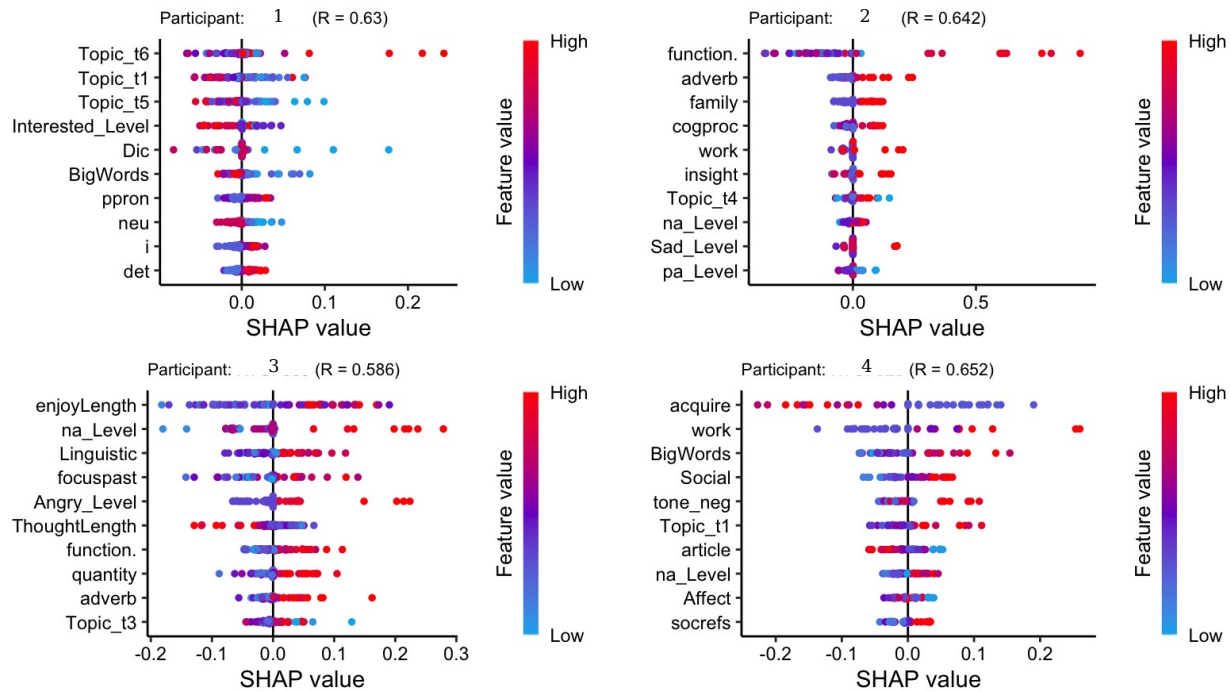
Note. Panel (a) Random Forest (b) Elastic Net. Each colored line represents an individual participant's predicted estimates (y axis) across different levels of actual negative affect (x axis).

Figure 3

Examples of person-specific (idiographic) predictions of negative affect for high-performance (top panel) and low-performance (lower panel) models



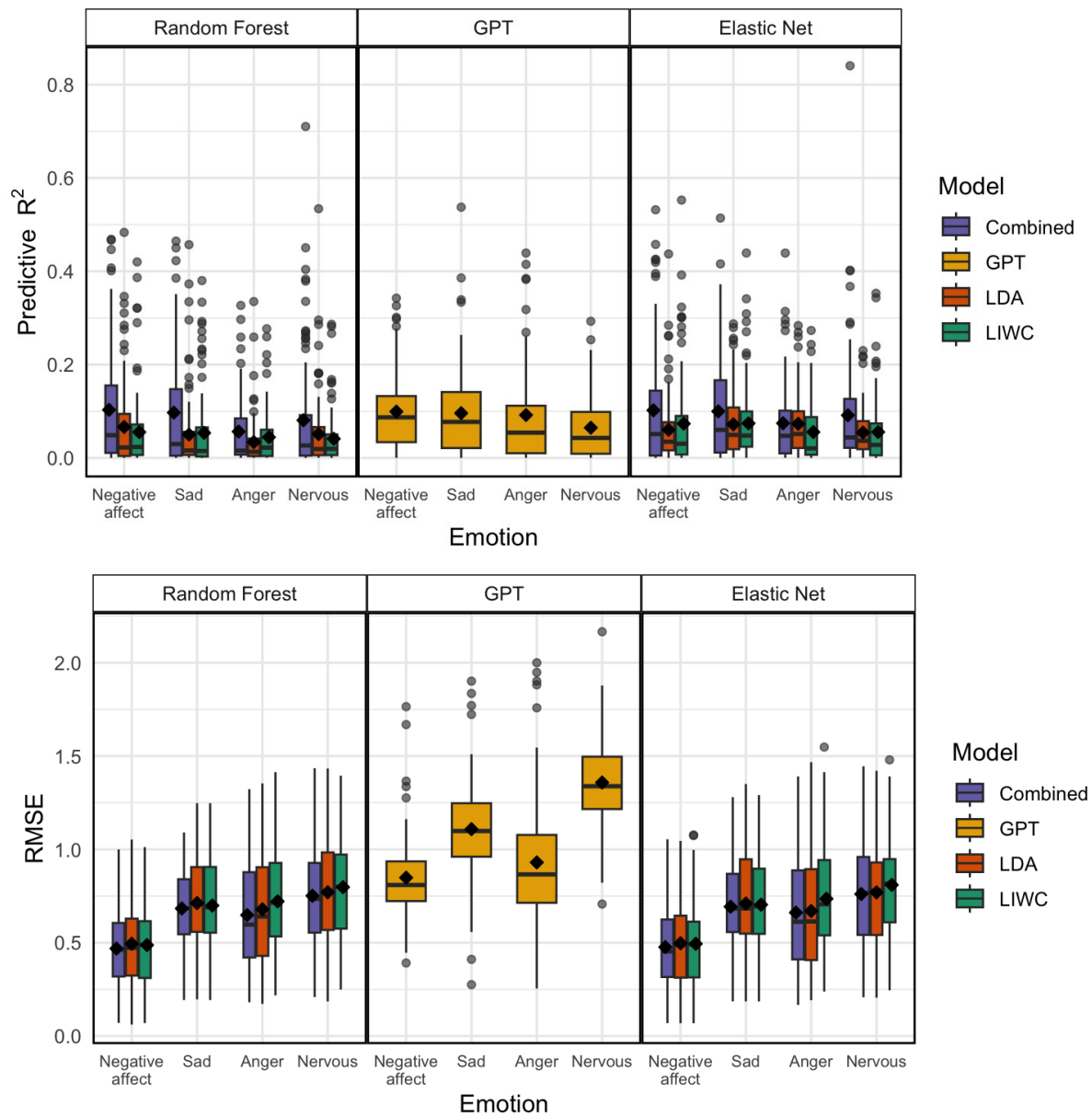
812 **Figure 4**
 813 SHAP (Shapley Additive Explanations) beeswarm plots for four individual participants,
 814 illustrating the contribution of different text-based features to the model's emotion predictions
 815
 816



817
 818 *Note.* Each dot represents an individual data point, with red indicating high feature values and blue indicating low feature values. The spread of dots across the x-axis reflects the variability in a feature's effect on predictions across different measurement time point. The y-axis lists the features in descending order of importance, meaning the top features had the strongest impact on the model's predictions. The x-axis represents SHAP values, indicating the magnitude and direction of each feature's impact on the predicted emotion scores. For example, for participant K23528, a *low* score on "Acquire" (red color) increases the model's prediction of negative affect. Conversely, a *high* feature value for "Work" or "Big Words" (red color) contributes to a higher predicted negative affect.

Figure 5

Performance metrics (R^2 and RMSE) for each NLP approach used in the idiographic models



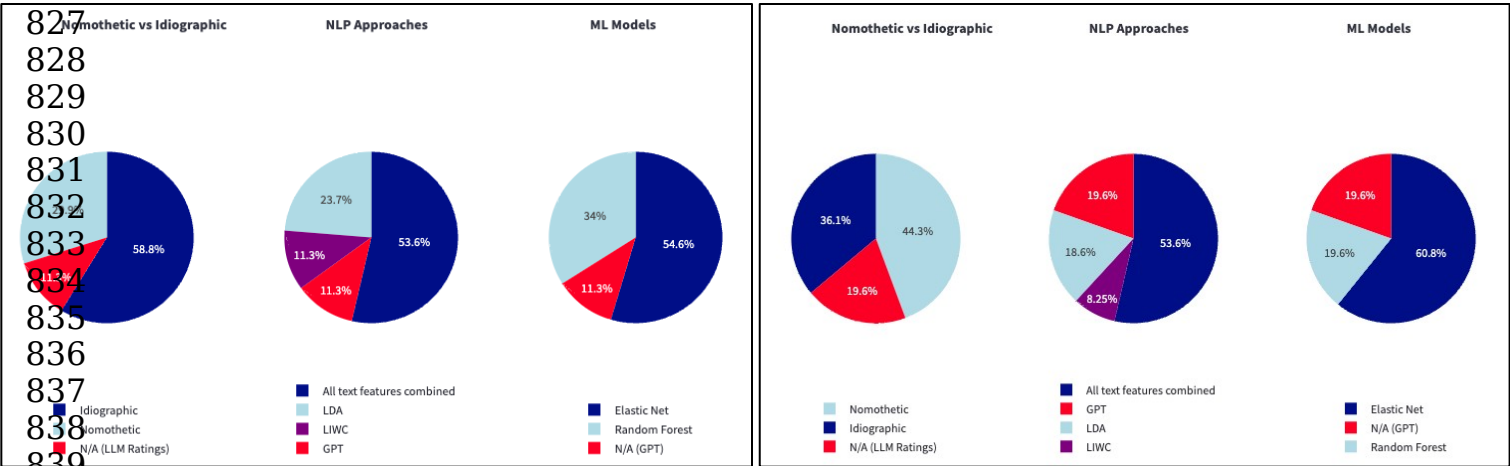
Note. The top panel displays the predictive R^2 for different models (Random Forest, GPT, and Elastic Net) across four emotion categories: negative affect, sadness, anger, and nervousness. The bottom panel presents the corresponding RMSE values. Each box plot represents the distribution of performance metrics for four different NLP-based feature sets: Combined (purple), GPT (yellow), LDA (orange), and LIWC+VADER (green). Higher R^2 indicates better predictive accuracy, while lower RMSE reflects better model fit.

822 **Figure 6**

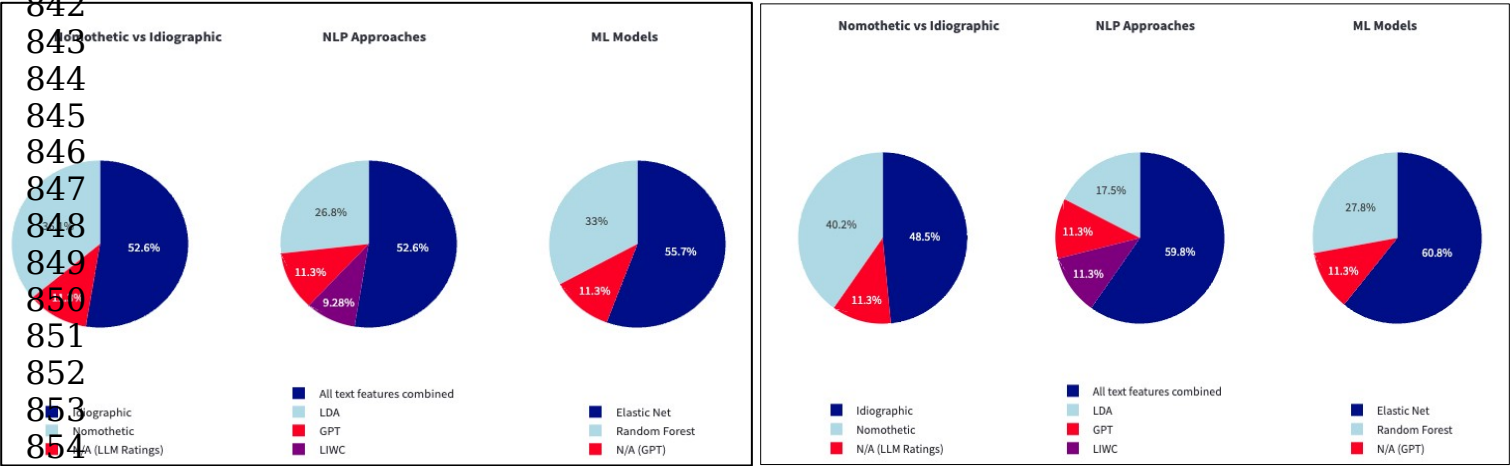
823 Distribution of best-performing models for predicting negative emotions across nomothetic vs.
824 idiographic frameworks, NLP approaches, and ML models

825 a) Negative Affect

b) Sadness



d) Nervousness



Note. The colors represent different model types, indicating the proportion of participants for whom each model was the best predictor.

